

The Mordell-Weil Theorem

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Abstract. The main objective of this project is to get an idea of the proof of a generalization of the Mordell theorem stating that the group $E(\mathbf{Q})$ of rational points of the elliptic curve E is a finitely-generated abelian group. Indeed, we will prove the Mordell-Weil's theorem (replacing the elliptic curve by an abelian variety A and $\mathbf{Q}$ by a number field K). The two main references are [BG06] and [HS00]. We first define and recall common objects from algebraic geometry and arithmetic geometry (Section 2). Then the Mordell-Weil theorem is proved (Section 3) and we conclude on some numerical computations (Section 4). This document is a description of what was done during this Twoples project under the direction of Preston Kelley from Oklahoma State University.

Disclaimer: content from Section 2 is not intended to be read too seriously. It is a quick presentation of the algebraic and arithmetic geometry machineries used. Before this project, the author knew barely nothing about elliptic curves, algebraic and arithmetic geometry. The author would be interested to pursue in a direction more oriented towards number theory.

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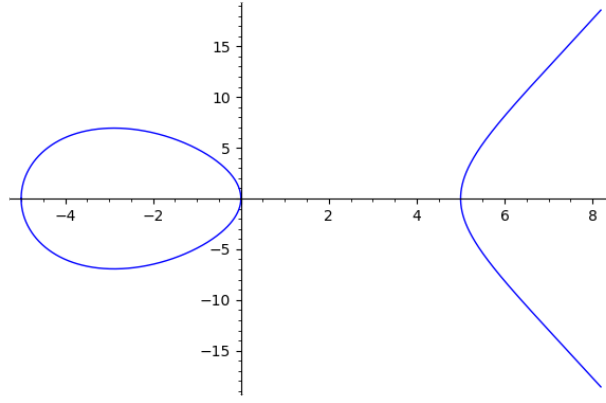
We first consider the case of a given elliptic curve E and see that the set of its rational points $E(\mathbf{Q})$ form a group with a very particular structure. The other sections are devoted to study a generalization of this example.

1 A toy example: $y^2 = x^3 - D^2x$

Let D be a square-free odd integer. The aim is to study the elliptic curve

$$E : y^2 = x^3 - D^2x$$

and its solution set $E(\mathbf{Q})$ composed of the pairs $(x, y) \in \mathbf{Q}_+ \times \mathbf{Q}$ satisfying the equation $y^2 = x^3 - D^2x$ (representing the elliptic curve C). We give hints to show that the Mordell-Weil group, i.e. the rational points of C , is a finitely generated abelian group, namely $\mathbf{Z} \oplus \mathbf{Z}/2\mathbf{Z} \oplus \mathbf{Z}/2\mathbf{Z}$. This is based on Ulm-Lyon 2003. When $D = 5$, the curve can be represented graphically.



Consider the homogenized equation of E in $\mathbb{P}^2$, i.e. $Y^2Z = X^3 - D^2XZ^2$. Basic manipulations with SageMath give that the rank is equal to 1 and that the points of the torsion subgroup are (in the projective plane) :

$$(0 : 1 : 0), \quad (5 : 0 : 1), \quad (-5 : 0 : 1), \quad (0 : 0 : 1).$$

For later considerations, note that C is an abelian subvariety of dimension 1 of $J_0(800)$. For more information, see the associated [LMFDB](#) page, *Elliptic curve with Cremona label 800a1*.

The proof is subdivided in two parts : a finiteness property and an implication. Proving the finiteness property on a specific function measuring the *complexity* of the elements constructed over an abelian group gives that finitely-generated modulo 2 implies finitely-generated. The finiteness property is being *admissible* and the specific function is a *height function*. These concepts are introduced hereafter. The overall proof for any elliptic curve E over $\mathbf{Q}$ is here barely sketched.

Sketch of the proof. Let $(\Gamma, +)$ be an abelian group where $\Gamma = \mathbf{Z}^r \oplus G$ with $r \in \mathbf{N}$ and G finite. Recall that this group is said to be finitely generated if there exists $r \in \mathbf{N}$ and $x_1, \dots, x_r \in \Gamma$ such that every $x \in \Gamma$ can be written as $\sum_{i=1}^r n_i x_i$ where the n_i are integers. The group Γ is finitely-generated modulo 2 if there exists a finite subset Z of Γ such that every element of Γ can be written as $z + 2y$ where $z \in Z$ and $y \in \Gamma$. A height function on Γ is a map $h : \Gamma \rightarrow \mathbf{R}_+$ for which there exists $M \geq 0$ such that

$$|h(x + y) + h(x - y) - 2h(x) - 2h(y)| \leq M \quad (1)$$

for every $(x, y) \in \Gamma^2$. A height is admissible if, for every $B \geq 0$, the set of $x \in \Gamma$ such that $h(x) \leq B$ is finite. Let h be a height on Γ . An admissible height $\hat{h}$ on Γ can be constructed from h . Indeed, define $\hat{h}(x)$ as the limit of $4^{-n}h(2^n x)$ (it is a Cauchy sequence).

Endowed with $\hat{h}$ and assuming Γ is finitely-generated modulo 2, the group Γ is finitely-generated. This statement will be proved with more generality in Section 3.

Take $\Gamma = E(\mathbf{Q})$, this is indeed a group by classical results from the theory of elliptic curves. It now *suffices* to show it is finitely-generated modulo 2 and endowed with an admissible height. This is done by (basic but tricky) arithmetic considerations and using arguments from the theory of valuations. $\square$

The aim of these notes is to bypass the (lengthy) computations with algebraic geometry and arithmetic geometry machinery. See [Col11, Appendix H.14] for the details of the proof (with lengthy computations) and especially see what happens in the case of a specific elliptic curve such as E . Note that, when $D = 5$, one can take M from Equation 1 to be equal to $6 \log 2000$.

As a general matter of fact, it is really hard to determine computationally the Mordell-Weil group. See for instance Section 6 from [GP25] for quite involved examples (genus 2 curves over the rational field $\mathbf{Q}$ and in particular the curve given by $y^2 = x^6 + x^4 + x^2 + 1$). For the computation of the Mordell-Weil group of a related curve, see [Sil09, Section X.6].

2 Algebraic and arithmetic geometry background

For the proof of the Mordell-Weil theorem (see Section 3), we do not need that much algebraic and arithmetic geometry. However an important part of the Twoples project was devoted to learn about classical objects and constructions in these fields (and to no longer be afraid of schemes). We recall some of them here and focus in particular on concepts needed for the proof of Mordell-Weil. The following pages are very condensed because they are not the heart of the project, despite being crucial. For complete references, see the bibliography.

2.1 From varieties to schemes

Let K be a field and $\overline{K}$ be an algebraic closure of K . The affine n -space $\mathbb{A}_K^n$ is equal to $\overline{K}^n$ endowed with the Zariski topology whereas the projective n -space is denoted by $\mathbb{P}_K^n$.

A subset Y is closed for this topology if and only if there exists $T \subset K[x_1, \dots, x_n]$ such that $Y = Z(T)$ with $Z(T) := \{\alpha \in \overline{K}^n : f(\alpha) = 0 \text{ for all } f \in T\}$. Then, a subset is closed if and only if it is an algebraic set, i.e. it consists of all common zeros of a finite number of polynomials with coefficients in K [SR13a]. The Zariski topology depends on K . For the projective space the Zariski topology consists of all common zeros of a finite number of homogeneous polynomials with coefficients in K .

An affine variety S is an irreducible algebraic subset of $\mathbb{A}^n$ for the Zariski topology, i.e. S is non-empty and cannot be written as the union of two proper closed subsets of $\mathbb{A}^n$. A projective variety is a projective irreducible algebraic set. We would like to define geometrically a similar notion when K is not necessarily a field but a mere ring. The main objective of this section is to understand the shift from the previous definitions to the one of a variety in the scheme-theoretic setting. We follow [SR13b].

Let A be a ring and denote by $\text{Spec}(A)$ the set of prime ideals of A (A is excluded from the points of $\text{Spec}(A)$).

Example 1. For instance, if $A = \mathbf{Z}$, the set $\text{Spec}(A)$ contains the usual prime ideals (p) for p a prime number and (0) . It is funny to compute $\text{Spec}(\mathbf{Z}^n)$. The prime ideals of $\mathbf{C}[x]$ and $\mathbf{R}[x]$ are somewhat easily determined. More generally, for a field K , $\text{Spec}(K[x])$ is made of the null ideal (0) and (P) for P an irreducible polynomial with coefficients in K . This is harder for $\mathbf{Z}[x]$. See Mumford's textbook [AM99] and [Bel15] for a picture of $\text{Spec}(\mathbf{Z}[x])$. For other interesting examples, see [Sta25, Section 10.27]. More geometrical situations can be thought of. An example is given by the spectrum of $\mathcal{O}_x$ being the local ring of a nonsingular point x of an algebraic curve. It is constituted of (0) and its unique maximal ideal $\mathfrak{m}_x$.

Note that the spectrum of a ring A is until then only a set. It can be endowed with the Zariski topology by defining the closed sets $V(E)$ as $\{\mathfrak{p} \in \text{Spec}(A) : E \subset \mathfrak{p}\}$ where $E \subset A$. These closed sets are obvious generalizations of the usual one on algebraic variety. Indeed, let $E \subset A$ and define $f_E(p)$ to be the image of E in A/p . Then $V(E) = f_E^{-1}(0)$.

Morally speaking, a scheme is no more than a space locally isomorphic to the spectrum of a ring. More formally, define a ringed space (*espace annelé*) as a pair $(X, \mathcal{O}_X)$ consisting of a topological space X and a sheaf of rings $\mathcal{O}_X$ called the structure sheaf. Then, a scheme is a ringed space $(X, \mathcal{O}_X)$ for which every point has a neighbourhood U such that the ringed space $(U, \mathcal{O}_X|_U)$ is isomorphic to the spectrum of a ring. See [SR13b, Section 5] for more information.

2.2 Divisors

Can we find an object determining uniquely every elements of the ring of polynomial functions on a variety? For polynomials in one variable, it would be the set of roots. For a polynomial in several variables, it would be hypersurfaces counted with multiplicities. In a general setting,

especially when unique factorization is not guaranteed, what could be such object? We are looking for an object that captures the multiplicity of zeroes and poles. This object is a divisor. It is defined as a finite formal sum of irreducible varieties of codimension 1. After introducing divisors, it would have been interesting to make explicit the link with line bundles.

Recall that codimension 1 subvarieties of curves are points. Codimension 1 subvarieties of surfaces are curves and so on. It is easy to think about the formal sum of points. For higher dimensional codimension 1 subvarieties, an arithmetic of these objects needs to be developed. Let $D = \sum n_i Y_i$ be a divisor where $n_i \in \mathbf{Z}$ and Y are codimension 1 subvarieties of X a fixed variety. We denote by $\text{Div}(X)$ the *group* of Weil divisors of X . By choosing adequately the n_i , we can replicate ideas from complex analysis and attach to a function f of $K(X)^*$ a divisor. Let $\text{ord}_Y(f)$ be the order of f on Y and define $\text{div}(f) = \sum_{\text{prime } Y} \text{ord}_Y(f) Y$ where a 1-codimensional subvariety Y is said to be prime if it is irreducible and closed. For non-zero elements, we obviously have $\text{div}(fg) = \text{div}(f) + \text{div}(g)$. A divisor D is said to be principal if there exists an f such that $D = \text{div}(f)$. This allows the definition of an equivalence relation the following way: $D, D' \in \text{Div}(X)$ are said to be linearly equivalent if their difference is a principal divisor. Two of the most familiar principal divisors are

$$(f)_0 = \sum_{\text{ord}_Y(f) > 0} \text{ord}_Y(f) Y \quad \text{and} \quad (f)_\infty = \sum_{\text{ord}_Y(f) < 0} \text{ord}_Y(f) Y$$

the divisor of zeros of f and the divisor of poles of f . Thus $\text{div}(f) = (f)_0 - (f)_\infty$. Quotienting out $\text{Div}(X)$ by the principal divisors, we obtain the divisor class group $\text{Cl}(X)$.

Example 2. For instance, $\text{Cl}(\mathbb{P}^n) \cong \mathbf{Z}$ since a divisor of $\mathbb{P}^n$ is principal if and only if it has degree 0 [HS00, Proposition A.2.1.3].

Note that, if the variety X is not necessarily regular, a generalization of Weil divisors exists: the Cartier divisors. See [BG06, A.8.17]. For our purpose, it is sufficient to get a feeling about Weil divisors except when one considers the pullback of the multiplication-by- m map (which we don't explicitly use hereafter apart from Section 2.5). See Section 2.5 for the definition of symmetric and anti-symmetric divisors.

Introduce the Riemann-Roch space of a divisor D as

$$\mathcal{L}(D) = H^0(X, \mathcal{O}_X(D)) = \{f \in \overline{K}(X) \mid \text{div}(f)D \geq 0\}.$$

Let $\{f_0, f_1, \dots, f_n\}$ be a basis of $\mathcal{L}(D)$, i.e. $\ell(D) := \dim_{\overline{K}} \mathcal{L}(D) = n + 1$. Suppose $\ell(D) \geq 1$. The divisor D is said to be very ample if the rational map $[f_0, f_1, \dots, f_n] : X \rightarrow \mathbb{P}^{\ell(D)-1}$ is an embedding, i.e. an isomorphism onto its image. The divisor D is ample if some multiple mD is very ample [Sil09].

Example 3. We compute $\mathcal{L}(D_F)$ where $D_F = \{F = 0\} \in \text{Div}(\mathbb{P}^n)$ for F a non-zero homogeneous polynomial of degree $d \geq 1$. By definition, we have that $\mathcal{L}(D_F)$ is constituted by all homogeneous polynomials of degree d in $x_0, \dots, x_n$. A basis is immediately given by the monomials of degree d , $x_0^{i_0} \dots x_n^{i_n}$, such that $i_0 + \dots + i_n = d$. The dimension of $\mathcal{L}(D_F)$ is thus given by the number of such monomials, i.e. number of ways to choose d elements among $n + d$ elements, i.e. $\binom{n+d}{d}$.

2.3 Heights

Before considering number fields, restrict to $\mathbf{Q}$. Define a place as an equivalence class of non-trivial absolute values. Denote by $M_{\mathbf{Q}}$ the set of non-trivial absolute values on $\mathbf{Q}$, i.e. the set of its places. It consists of an archimedean one $|\cdot|_\infty$ and one non-archimedean $|\cdot|_p$ for each prime number p . Recall that two absolute values are equivalent if they define the same

topology. A more concrete condition gives the equivalence of $|\cdot|_a$ and $|\cdot|_b$ if and only if there exists a positive real number s such that $|\cdot|_a = |\cdot|_b^s$. Of course, non-trivial archimedean and non-archimedean absolute values can not be equivalent. Thus $\mathbf{Q}$ has a place at infinity and one for each prime. For an arbitrary number field K , the archimedean places are given by the real embeddings of K and by the complex conjugates pairs of embeddings of K .

Example 4. For instance, the two embeddings of $\mathbf{Q}[\sqrt{2}]$ are given by $a + b\sqrt{2} \mapsto a + b\sqrt{2}$ and $a + b\sqrt{2} \mapsto a - b\sqrt{2}$. Thus v_1 is given by $|a + b\sqrt{2}|$ and v_2 is given by $|a - b\sqrt{2}|$. It is easy to see how Galois theory comes into the game and that non-archimedean places are associated with non-zero prime ideals. For precise details, see [HS00, Proposition B.1.3]. See [BG06, Example 1.3.9] to determine all extensions of an absolute value of $\mathbf{Q}$ to $\mathbf{Q}(\zeta)$ with ζ a primitive p -th root of unity for p an odd prime number.

With no surprise, M_K is the set of absolute values on a number field K . A height on algebraic points of $\mathbb{P}_{\mathbf{Q}}^n$ measures the "algebraic complications" needed to describe a point [BG06]. We define

$$h(\mathbf{x}) := \sum_{v \in M_K} \max_j \log |x_j|_v$$

where $\mathbf{x} = (x_0 : \dots : x_n) \in \mathbb{P}^n(\overline{\mathbf{Q}})$. This application is well-defined. Indeed, it is independent of the choice of K [BG06, Lemma 1.5.2] and it gives the same result for an element collinear with x by the product formula¹. It is the absolute logarithmic height of the point represented by $\mathbf{x}$. Note that we use the following normalization $|p|_p = 1/p$ for p a prime and otherwise $|\cdot|_\infty$ is the archimedean absolute value. Extending to number fields is possible but requires different normalisations. For the general setting, see [BG06, p. 11]. By embedding $\mathbb{A}_{\mathbf{Q}}^n$ in $\mathbb{P}_{\mathbf{Q}}^n$ by restricting the zero-th coordinate of $\mathbf{x}$ to be 1, one defines a height function in the affine case. Indeed :

$$h((1 : x_1 : \dots : x_n)) = \sum_{v \in M_K} \max_j \{0, \log |x_1|_v, \dots, \log |x_n|_v\}. \quad (2)$$

Remark 5. From Equation 2, one can talk about the height of an algebraic number. See [BG06, Theorem 1.5.9] for a proof of the Kronecker's theorem stating that the height of a non-zero algebraic number ζ is zero if and only if ζ is a root of unity. See [HS00, Corollary B.2.3.1] for a projective version of Kronecker's theorem. A more general statement also exist for abelian varieties equipped with a very specific height function where we replace the group of roots of unity by the torsion points (see Section 2.5). It is more interesting to prove the Kronecker theorem (at least in the basic setting) using the fact that $h(\alpha^n) = |n|h(\alpha)$ for any $n \in \mathbf{Z}$ and $\alpha \in \overline{\mathbf{Q}}$ and that there are only finitely many algebraic numbers of bounded degree and bounded height. This last statement is known as Northcott's theorem, see Theorem 6.

The definition of a height function can then be generalized to polynomials (and the height function is related to the Mahler measure but that is sadly not of particular interest to us, except to prove Northcott's theorem). Then, the definition of height extends from projective space to any projective variety X defined over a number field K by using the geometry of the variety. Let D be a Cartier divisor on X . If D is very ample, it induces a closed embedding $\phi_D : X \hookrightarrow \mathbb{P}_{\overline{K}}^N$. We define the height of a point $P \in X(\overline{K})$ relative to the divisor D as the standard height of its image

$$h_D(P) = h(\phi_D(P)).$$

¹Just to mention them at least, product formulas are really central and useful but sadly we don't need them that much for this project.

In the case where the Cartier divisor D is only ample, there exists an integer $m \geq 1$ such that the multiple mD is very ample. The height is then defined by linearity

$$h_D(P) = \frac{1}{m} h_{mD}(P).$$

This defines the height function $h_D : X(\overline{K}) \rightarrow \mathbb{R}$ uniquely up to a bounded function. The association $D \mapsto h_D$ is linear (modulo bounded functions).

We conclude by stating the Northcott's theorem for projective varieties over number fields. This theorem allows to prove that a set of point of bounded height (and degree) is indeed finite. That is one of the main interest of height functions: prove finiteness.

Theorem 6. *Let X be a projective variety defined over a number field K and let h_D be a height function associated to an ample divisor D on X . Then the set*

$$\{P \in X(\overline{K}) \mid h_D(P) \leq C, [K(P) : K] \leq d\}$$

is finite for any positive constant d and real C .

Proof. Without loss of generality (see [BG06, Theorem 2.4.9]), it is sufficient to prove the theorem for $K = \mathbf{Q}$ and h_D the standard height on $\mathbf{P}_{\mathbf{Q}}^1$. Thus, it is sufficient to prove that there are only finitely many algebraic integers of bounded height H and degree d . Let $\alpha \in \overline{\mathbf{Q}}$ and let $P(x) \in \mathbf{Z}[x]$ be the minimal polynomial of α . Write it as

$$P(x) = a_n x^n + a_{n-1} x^{n-1} + \dots a_2 x^2 + a_1 x + a_0$$

where $a_n \neq 0$ and $n \leq d$. The coefficients of P can be bounded by $(2C)^d$ by classical root-coefficient inequalities and the binomial theorem. Since there are $n+1$ coefficients for each P , this bounds the number of distinct polynomials P . Since each of these polynomials has n roots, the number of algebraic integers of height at most C and degree d is then finite. $\square$

2.4 Elliptic curves and abelian varieties

Of main interest for us is the genus of a curve. It would be too demanding to fully understand the definition of the arithmetic genus (see [Sta25]): let K be a field, let X be a proper scheme over K having dimension 1 and $H^0(X, \mathcal{O}_X) = K$, then the (arithmetic) genus of X is $g = \dim_K H^1(X, \mathcal{O}_X)$. Still a little bit demanding but not too much, we refer to the Riemann-Roch theorem to define the genus of elliptic curves. In this section, we first restrict to curves of genus 1. A curve of genus ≥ 2 is called an hyperelliptic curve (if it is a double covering of the projective line). Afterwards, will be introduced a generalization of elliptic curves, namely abelian varieties.

An elliptic curve over a field K is a projective non-singular curve of genus 1 with a K -rational point $\mathcal{O}$. Fortunately enough, elliptic curves can be described by equation (but abelian varieties can not in general). In the projective setting, the Weierstrass equation of an elliptic curve E over a field K is given by

$$y^2 z + a_1 x y z + a_3 y z^2 = x^3 + a_2 x^2 z + a_4 x z^2 + a_6 z^3$$

where a_1, a_2, a_3, a_4, a_6 and $\Delta \neq 0$. The discriminant Δ of E is defined such that $\Delta := -b_2^2 b_8 - 8b_4^3 - 27b_6^2 + 9b_2 b_4 b_6$ where $b_2 = a_1^2 + 4a_2$, $b_4 = 2a_4 + a_1 a_3$, $b_6 = a_3^2 + 4a_6$, $b_8 = a_1^2 a_6 + 4a_2 a_6 - a_1 a_3 a_4 + a_2 a_3^2 - a_4^2$. When K is not of characteristic 2 or 3, the Weierstrass equation can be reduced to

$$y^2 z = x^3 + a x z^2 + b z^3$$

where $\Delta := 4a^3 + 27b^2 \neq 0$.

Let E be an elliptic curve over a field K given by the equation $f(x, y, z) = 0$. The set of K -rational points $\{(x, y, z) \in \mathbf{P}^2(K) \mid f(x, y, z) = 0\}$ is denoted by $E(K)$. At first sight, it is not trivial that an elliptic curve can be endowed with a group law. It is even less for $E(K)$. Recall that one of the aim of this project is to describe the structure of $E(K)$, see Theorem 7.

Elliptic curves can be generalized in the following natural way. A group variety V is a variety which can be endowed with the usual group morphisms, i.e. the multiplication $m : V \times V \rightarrow V, (x, y) \mapsto xy$, the inverse $\iota : V \rightarrow V, x \mapsto x^{-1}$ and the identity e . An abelian variety is then no more than a projective group variety.² It is necessarily a smooth projective variety (by translating a non-singular point and reaching every point of the variety; the proof is not as trivial as one might think, see [Dar22, Proposition 2.7]).

2.5 Canonical heights on abelian varieties

The main reference for this section are Silverman's lecture notes for the Arizona Winter School and [BG06]. Let A be an abelian variety over a number field K and D a Cartier divisor on A . The divisor D is said to be symmetric if $[-1]^*D \sim D$, i.e. they are linearly equivalent. The divisor D is said to be antisymmetric if $[-1]^*D \sim -D$. Suppose now that D is either symmetric or antisymmetric and let ρ be a constant equal to 2 if D is symmetric and 1 if it is antisymmetric. The following results, that are parts of the Neron-Tate theorem, generalizes ideas and results from Section 1. First, the following limit converges

$$\widehat{h}_D(P) = \lim_{n \rightarrow \infty} \frac{h_D([2^n](P))}{2^{\rho n}}.$$

The proof of that last fact is a nice analysis argument. This is a simple argument ending up being really powerful.

Proof. One needs to prove it is a Cauchy sequence since $\mathbf{R}$ is complete. The proof goes as follow. Let $a_n = h_D([2^n](P))/2^{\rho n}$. Note that $a_{n+1} - a_n$ can be bounded by $C_1/2^{\rho(n+1)}$ by the height bounding property [BG06, Proposition 1.5.15]. For $m \geq n$, we have

$$a_m - a_n = (a_m - a_{m-1}) + (a_{m-1} - a_{m-2}) + \cdots + (a_{n+1} - a_n).$$

Then, by the triangle inequality, we get $|a_m - a_n| = \sum_{k=n}^{m-1} |a_{k+1} - a_k|$. We conclude as it can be bounded by the tail of a convergent (geometric) series. $\square$

Then, there exists a constant C which depends on A and D such that

$$\left| \widehat{h}_D(P) - h_D(P) \right| \leq C \quad \text{for all } P \in A(\overline{K}).$$

Moreover, the following property is relevant for the proof of the Mordell-Weil theorem. For all $P, Q \in A(\overline{K})$, the height function satisfies the parallelogram law

$$\widehat{h}_D(P + Q) + \widehat{h}_D(P - Q) = 2\widehat{h}_D(P) + 2\widehat{h}_D(Q).$$

Using the parallelogram law, one observes that $\widehat{h}_D$ is a quadratic form.

To conclude, note that Northcott's theorem can be applied to the canonical height.

²More formally, an abelian variety is a geometrically irreducible and geometrically reduced complete group variety [BG06]. See [HS00] or [Mil08] for a detailed overview of abelian varieties over $\mathbf{C}$. More generally, an abelian variety over k is a group scheme over $\text{Spec}(k)$ which is proper (complete, separated and of finite type) and geometrically integral.

3 The Mordell-Weil theorem

We now prove the Mordell-Weil theorem giving the structure of rational points on a (generalization of an) elliptic curve.

Theorem 7. *Let K be a number field and A an abelian variety over K . The K -rational points of A form a finitely-generated abelian group denoted by $A(K)$.*

Note that this theorem can be extended to K a global field [Gar20]. When K is restricted to be $\mathbf{Q}$ and A is an elliptic curve E , Theorem 7 gives a decomposition of $E(\mathbf{Q})$ as follows

$$E(\mathbf{Q}) = \mathbf{Z}^r \oplus E(\mathbf{Q})_{\text{tors}}$$

where r is the rank of E and $E(\mathbf{Q})_{\text{tors}}$ is the torsion part.

Example 8. *For instance, when $K = \mathbf{Q}$, the torsion part $E(\mathbf{Q})_{\text{tors}}$ is described by Mazur's torsion theorem. Some examples are computed in [Sil99, p. 88] and [Rog00]. For instance, we have*

$$\begin{array}{ll} E : y^2 = x^3 - 2x, & E(\mathbf{Q}) = \mathbf{Z} \oplus \mathbf{Z}/2\mathbf{Z}, \\ E : y^2 = x^3 + 17, & E(\mathbf{Q}) = \mathbf{Z}^2, \\ E : y^2 = x^3 - 17x, & E(\mathbf{Q}) = \mathbf{Z}^2 \oplus \mathbf{Z}/2\mathbf{Z}, \\ E : y^2 = x^3 - (2 \cdot 7 \cdot 11)^2 x, & E(\mathbf{Q}) = \mathbf{Z}^2 \oplus \mathbf{Z}/2\mathbf{Z} \oplus \mathbf{Z}/2\mathbf{Z}, \\ E : y^2 = x^3 - (2 \cdot 3 \cdot 5 \cdot 11 \cdot 19 \cdot 41 \cdot 43 \cdot 67 \cdot 83)^2 x, & E(\mathbf{Q}) = \mathbf{Z}^6 \oplus \mathbf{Z}/2\mathbf{Z} \oplus \mathbf{Z}/2\mathbf{Z}. \end{array}$$

Note that the second example does not contradict Theorem 1.3 from [FN10]. For more examples, see LMFDB. The result can vary with respect to the number field, see

$$E : y^2 = x^3 - x, \quad E(\mathbf{Q}) = \mathbf{Z}/2\mathbf{Z} \oplus \mathbf{Z}/2\mathbf{Z}, \quad E(\mathbf{Q}[i]) = \mathbf{Z}/2\mathbf{Z} \oplus \mathbf{Z}/4\mathbf{Z}.$$

More generally, Theorem 7 gives the following decomposition

$$A(K) = \mathbf{Z}^r \oplus A(K)_{\text{tors}}$$

for K a number field and A an abelian variety. As usual, r is the rank.

The proof of Theorem 7 follows closely the sketch given in Section 1 (where A is an elliptic curve and K is $\mathbf{Q}$). We first (re-)sketch it more generally. The proof is decomposed in two main steps: the weak Mordell-Weil (Section 3.1) and then concluding (Section 3.2). We follow the classical references [SB90], [HS00] and [BG06].

3.1 Weak Mordell-Weil and descent lemma

We prove the two main results needed to prove the Mordell-Weil theorem. It seems that the *correct* setting to consider is Galois cohomology. We don't. Moreover, we assume some classical results (to be found in [BG06] for instance).

Theorem 9. *Let K be a number field and A an abelian variety over K . Let m be a positive integer. Then $A(K)/mA(K)$ is finite.*

Proof. Let $A[m]$ denote the group of m -torsion points of $A(\bar{K})$. According to [BG06, Section 10.4], we may assume without loss of generality that $A[m] \subset A(K)$. It is highly interesting to consider the field extension $L = K(\frac{1}{m}A(K))$ generated by the coordinates of all points $Q \in A(\bar{K})$ such that mQ is K -rational. By the Chevalley and Weil theorem (applied to the multiplication-by- m map, see [BG06, Section 10.3]), the extension L/K is unramified outside a

finite set of places of K . Consequently, by the Hermite-Minkowski theorem (see [SB90, Section 4.1]), the extension L/K is finite.

Since $A[m] \subset A(K)$, standard Kummer theory (see Section [BG06]) yields a bilinear pairing:

$$\langle \cdot, \cdot \rangle : (A(K)/mA(K)) \times \text{Gal}(L/K) \rightarrow A[m],$$

defined by $\langle P, \sigma \rangle = \sigma(Q) - Q$ where $P = mQ$. This pairing is non-degenerate (has zero right and left kernels, see [BG06, p. 343]). Since $\text{Gal}(L/K)$ is finite (as shown above) and $A[m]$ is finite, it follows that $A(K)/mA(K)$ must be finite. $\square$

Theorem 10. *Let $A(K)$ be the abelian group of K -rational points of A and let $m \geq 2$ be a positive integer. Let also $\| \cdot \| = \hat{h}_{\mathbf{c}}$ be the Neron-Tate height associated to $\mathbf{c}$ as introduced in Theorem 6 and defined in Section 2.5. Assume that S is a set of representatives for $A(K)/mA(K)$, bounded relatively to $\| \cdot \|$ by a constant C . For any $x \in A(K)$, there is a decomposition*

$$x = \sum_{i=0}^l m^i y_i + m^{l+1} z$$

where $y_i \in S$ and where $z \in A(K)$ satisfies $\|z\| \leq C + 1$. In particular, $A(K)$ is generated by elements in the ball $\{x \in A(K) : \|x\| \leq C + 1\}$.

The proof goes by induction the number of elements needed to decompose x . See [BG06, Lemma 10.6.2] for a complete description of it. See also [BG06, Section 9.2]. Note that, one can replace $\| \cdot \|$ by any quadratic form defined over G an arbitrary abelian group (see [HS00, Lemma C.0.3]).

3.2 End of the proof

Combining previous results, we can prove Theorem 7.

Proof. Let K be a number field and A an abelian variety over K . Let $m \geq 1$ be an integer. It is sufficient to consider $m = 2$ for instance. By the weak Mordell-Weil theorem (Theorem 9), we have that $A(K)/2A(K)$ is a finite group. We now apply the descent lemma (Theorem 10) to the Neron-Tate canonical height coming from the Weil height h associated to a symmetric ample divisor in order to prove that $A(K)$ is generated by a bounded set. From Northcott's theorem (Theorem 6), this is sufficient to show that $A(K)$ is finitely generated. $\square$

4 Numerical observation over fixed cubic fields

We use Sage to get a rough average estimate of the rank of elliptic curves over a fixed cubic field. Unfortunately, our program is too naive since it counts more than once some elliptic curves. We first define the number fields considered.

```
from sage.all import *

# Cubic Fields used for computations
x = polygen(QQ, 'x')
cubicFields = [NumberField(x**3 + x**2 + 997*x + 1, 'g_0'),
NumberField(x**3 - 2, 'g_1'),
NumberField(x**3 - 3, 'g_2'),
NumberField(x**3 - 5, 'g_3'),
NumberField(x**3 - 7, 'g_4'),
NumberField(x**3 - 13*x + 1, 'g_5')]
```

Then construct some arrays. One is especially useful because it contains all points less than a finite given height.

```
BOUND = Integer(10)

pointsHeight = []
possibleEmbeddings = [] # We have 3 roots!
possibleEmbeddingsVal = []

for numberfield in cubicFields:
    pointsHeight.append(list(numberfield.elements_of_bounded_height(bound=
                                                                    BOUND)))

    embNumField = [numberfield.embeddings(CC)[i] for i in range(0, 3)]
    numNumField = [val(numberfield.gen()) for val in embNumField]
    possibleEmbeddings.append(embNumField)
    possibleEmbeddingsVal.append(numNumField)

    numberfield.inject_variables()
```

Selecting a cubic extension of $\mathbf{Q}$, we compute a naive version of the average rank. Note that we might have some problems in the code since some elliptic curves could be counted more than one time.

```
K = cubicFields[1]
consideredPointsHeight = pointsHeight[1]

total_rank, nb_curve, counter = 0, 0, 0
processed_j_invariants = set()

for A in consideredPointsHeight:
    for B in consideredPointsHeight:
        try:

            # 1. Define the Elliptic Curve E:  $y^2 = x^3 + Ax + B$  over  $K = \mathbf{Q}\mathbf{Q}$ 
            E = EllipticCurve(K, [0, 0, 0, A, B])

            if E.discriminant() == 0: continue

            j = E.j_invariant()
            if j in processed_j_invariants: continue
            processed_j_invariants.add(j)

            r_lower, r_upper, _ = E.simon_two_descent()
            rank_estimate = r_lower # Use the lower bound as the estimate

            total_rank += rank_estimate
            nb_curve += 1

            if counter % 1 == 0: # Print every successful calculation
                average_rank = float(total_rank / nb_curve) if nb_curve > 0 else 0
                j_display = f"{j.n(5)}" if j.denominator() != 1 else str(j)
                print(f"{A}\t {B}\t | {j_display}\t | {rank_estimate}\t | {nb_curve}
                        \t\t | {average_rank:.4f}")

                counter += 1

        except Exception as e:
            print(f"Skipping E(A={A}, B={B}): Error during descent: {e}")
            continue
```

Note that we do not use a method to check if two elliptic curves are isomorphic since it was computationally too expensive. Nevertheless, from this code we might get a probable upper bound of the average rank (approximately 0.80 for $\mathbf{Q}[\sqrt[3]{2}]$). Proving such conjectures seems incredibly difficult. Even for $\mathbf{Q}$ a lot of problems arise. More generally, why is it so difficult to compute the rank over a number field K , even over $\mathbf{Q}$? First, because the rank is way more known conjecturally than certainly (it seems to be related to the behaviour of L-functions by the Birch and Swinnerton-Dyer Conjecture). Secondly, because it is related to the Tate-Shafarevich group for which there is no known, general, finite-time algorithm to determine if it is trivial. However, an upper bound on the rank can be given by the 2-Selmer group. Specifically for elliptic curves, if we could compute the group $E(\mathbf{Q})/2E(\mathbf{Q})$, then we could apply the descent procedure and have an effective method.

To conclude we give an overview of the difficulty to determine explicitly $E(\mathbf{Q})/2E(\mathbf{Q})$. Using the example from Section 1, recall that we consider $E : y^2 = x^3 - D^2x$. Even in that very specific example, one has to introduce the following non-trivial map $\phi : E(\mathbf{Q}) \rightarrow (\mathbf{Z}/2\mathbf{Z})^{2s+2}$ sending ∞ on $(0, \dots, 0)$ and $P = (x, y)$ on

$$\left(\overline{v_2(x)}, \overline{v_{p_1}(x)}, \dots, \overline{v_{p_s}(x)}, \overline{v_2(x+D)}, \overline{v_{p_1}(x+D)}, \dots, \overline{v_{p_s}(x+D)} \right)$$

where $v_p(\cdot)$ is the p -adic valuation and $s = \#\{p_1, \dots, p_s\}$ the cardinal of the set of prime divisors of D . Note that $\ker \phi = 2E(\mathbf{Q})$. Using that, one can prove that $E(\mathbf{Q})$ is finitely-generated modulo 2. This is thus absolutely not trivial to determine $E(K)/2E(K)$ in general, for K a number field. This is even more complicated for abelian varieties.

It would be really interesting to continue this project following [Bal+16].

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